Technology Stack

Date	2 July 2026
Team ID	
Project Name	**Voice based concept understanding analyser**
Maximum Marks	2 Marks

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.NO	ARCHITECTURE COMPONENT/LAYER	TECHNOLOGY CHOSEN	JUSTIFICATION / PURPOSE
1	**Frontend / Client-Side** (e.g., React, Angular, HTML/ CSS)	**React.js & Tailwind CSS**	React provides extreme component reusability and real-time state manipulation to handle audio recording states and render structural knowledge graphs seamlessly. Tailwind CSS provides clean utility-first styling for responsive dashboards across devices.
2	**Backend / Server-Side** (e.g., Node.js, Python/ Django, Java)	**Python (FastAPI)**	FastAPI is explicitly chosen for its exceptional performance speed and native support for asynchronous event-driven calls. This is vital for streaming concurrent heavy audio file buffers efficiently and interfacing natively with Python-based AI/NLP toolkits.
3	**Database / Data Storage** (e.g., MySQL, MongoDB, PostgreSQL)	**PostgreSQL & pgvector**	PostgreSQL offers enterprise-level data relational mapping for core tracking data, profiles, and logs. Utilizing the `pgvector` database extension allows the system to store and perform mathematical vector similarity operations directly on conceptual text embeddings inside database queries.

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4	**Cloud / Hosting / Deployment** (e.g., AWS, Azure, Docker)	**Docker & AWS (ECS / S3)**	Docker containers ensure absolute environmental consistency between local debugging and production. AWS ECS handles easy scalability of worker routines based on processing traffic, while Amazon S3 reliably handles cost-effective cloud object storage for transient raw audio audio wave records.
5	**Version Control & CI/ CD** (e.g., GitHub, GitLab, Jenkins)	**GitHub & GitHub Actions**	Enables seamless collaborative code management among team members. GitHub Actions is utilized for integrated automated testing, lint validation, and immediate continuous deployment triggers out to cloud instances upon structural core baseline merges.
6	**Third-Party APIs / Other Tools** (e.g., Stripe, Twilio, Google Maps)	**OpenAI Whisper API & Hugging Face**	The OpenAI Whisper API enables high-accuracy state-of-the-art automated speech-to-text voice transcriptions capable of adapting to various accents. Hugging Face libraries (Sentence-Transformers) generate dense semantic text embeddings to perform mathematical conceptual rubric validation.